Eight fixed points in the odd-degree orbit budget

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be transitive with a connected string C-marking and all maximal omissions intransitive. Assume $r \ge 6$ and $n = 2r + 1$. Retain the actual six-point head on $D = \{1, \dots, 6\}$ and the literal interface

$$\tau = \rho_3 = (6\ b)\delta, \quad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus \{1, \dots, 5\}), \quad \rho_4|_D = 1, \quad G_{\ge 5}|_{D \cup \{b\}} = 1,$$

where δ is supported outside $D \cup \{b\}$. Put $H = G_{\ge 5}$ and $K = G_{>5}$. Suppose every tail colour $j \ge 5$ has two full ambient-eligible edges in one actual H-orbit. Normalize these assignments by the acquired whole-orbit transport, so every assigned orbit has intransitive K-action. No assignment of an eligible pair split between distinct H-orbits is inferred.

Theorem 1. *Under these hypotheses, H cannot have exactly eight fixed points.*

Proof. Suppose it does. For each nontrivial H-orbit O_i, let F_i be its assigned colours and put

$$d_i = |F_i| - |O_i|/2.$$

Let p index the orbit assigned colour 5. Since the sets F_i partition all $r-5$ tail colours, the equation $n = 2r + 1$ gives

$$\sum_i d_i = (r - 5) - (n - 8)/2 = -3/2. \tag{1}$$

The acquired normalized assignment theorem gives

$$d_p \le 0, \qquad d_i \le -1 \text{ if } |O_i| \text{ is even}, \quad d_i \le -\tfrac{1}{2} \text{ if } |O_i| \text{ is odd} \quad (i \ne p). \tag{2}$$

An empty assignment has the exact deficit $-|O_i|/2$.

The odd involution δ centralizes H. It fixes the seven specified fixed points and therefore also the only eighth fixed point. We first show it preserves every nontrivial H-orbit. If it exchanges orbits O_i, O_j of equal size m, transport the assigned edges of O_j by the actual involution $\tau \in G_\ell$ for every tail colour ℓ. On these orbits its action is that of δ, and it centralizes H. Thus each colour in $F_i \cup F_j$ has two ambient-eligible edges in O_i. They are also eligible for the full local omission of $H|_{O_i}$, by subgroup inclusion. The complete selected colour bound gives

$$|F_i| + |F_j| \le \lfloor m/2 \rfloor, \qquad d_i + d_j \le -\lceil m/2 \rceil.$$

For $m \ge 3$ this is at most -2, contrary to (1)–(2). For $m = 2$, neither orbit can contain two disjoint edges of one involution, so both assigned sets are empty and the deficit is exactly -2, again impossible.

Because δ is odd and fixes the eight fixed points, it acts oddly on some preserved nontrivial orbit O. Its restriction there is a nonidentity involution centralizing a transitive group, hence is free. Write $|O| = 2q$, where q is odd.

First suppose $q \ge 3$. Its assignment cannot be empty, which would contribute $-q \le -3$ to (1). The acquired internal-pair forest gives $d_O \le -1$. If $O \ne O_p$, integrality and (1)–(2) force $d_O = -1$. The reviewed odd-degree row theorem applies: its assigned $q-1$ colours form a consecutive interval whose last label z is strictly below $r - 1$, and

$$|\operatorname{Fix}(G_{>z})| \ge 2z + 3.$$

Every colour after z retains its ambient eligible pair. The full terminal interval bound therefore gives

$$|\operatorname{supp}(G_{>z})| \geq 2(r-1-z)+1.$$

The interval is nonempty and is intransitive on Ω because it fixes the specified head. Adding the last two inequalities yields $n \geq 2r+2$, contrary to $n = 2r+1$.

If instead $O = O_p$, (1)–(2) and integrality give $d_p = -1$. The remaining total deficit is $-1/2$. There is therefore exactly one further nontrivial orbit V, of odd size, with $d_V = -1/2$. In particular F_V is nonempty; a nontrivial odd orbit with empty assignment would contribute at most $-3/2$. The internal equality theorem gives

$$F_p = \{5, \ldots, q+3\}, \qquad I = F_V = \{q+4, \ldots, r-1\}, \quad 2|I| = |V| - 1,$$

and every colour outside F_p has trivial image on the δ-pairs of O. Every possible edge of a colour in I on O is consequently one of these pairs, already joined by $\tau \in G_j$, and is ambient ineligible. The eight fixed points carry no tail edges. All eligible edges of colours in I are therefore confined to V. Since $T = G_I \leq K$ and K is intransitive on V, the acquired localization theorem gives

$$|V| - 1 = 2|I| \leq |V| - o_T(V) \leq |V| - 2,$$

a contradiction. This excludes $q \geq 3$ in every position.

Finally $q = 1$. No tail edge on the two-point orbit O is eligible, since its points are joined by τ in every tail omission. Its assignment is empty and $d_O = -1$. Only $-1/2$ of the total deficit remains. The involution δ cannot act nontrivially on any other nontrivial orbit W. Such an action would be free. For $|W| = 2$, the same argument contributes another -1. For $|W| \geq 4$, an empty assignment contributes at most -2; a nonempty assignment contributes at most -1 by the internal pair forest. Its elementary forest bound also holds for two pairs: at most one distinct selected colour can join the two pair vertices. Each possibility contradicts (1)–(2), including when $W = O_p$.

Thus δ is literally the one transposition on O. The reviewed odd-degree exterior-pair obstruction applies: all its actual head and natural-R hypotheses have been retained, and every colour $j \geq 6$ has an ambient eligible pair by the assignment. It excludes precisely this $\tau = (6\ b)\delta$ configuration for $n = 2r+1$. The final case is impossible, proving the theorem. □

Attribution and remaining scope. The normalized assignment and selected-colour inequalities, internal pair equality, complete localization proof, strict odd-degree row exit and exact exterior-pair obstruction are acquired inputs. The new step assembles their full parity and deficit cases at exactly eight fixed points, including exchanged orbits and an empty internal pair. This does not supply the assignment when eligible pairs split across different orbits. The seven-fixed-point budgets and the global separator argument remain. No alternating rank induction, new computation, formal declaration, original or question credit is claimed.